

Question:

Write a JAVA program to implement bank functionality in the above scenario and demonstrate the dynamic polymorphism concept. Note: Create few classes namely Customer, Account, RBI (Base Class) and few derived classes (SBI, ICICI, PNB etc). Assume and implement required member variables and functions in each class.

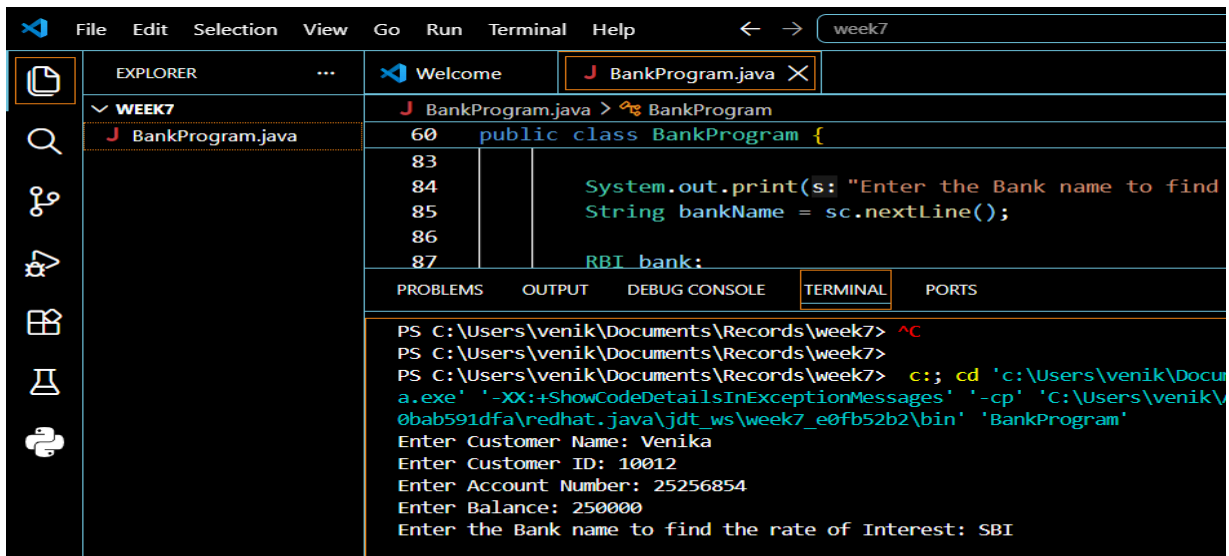
1. Import the `Scanner` class.
2. Create a `Customer` class to store customer name and customer ID.
3. Create an `Account` class to store account number and balance.
4. Create an `RBI` base class with a minimum interest rate of 4%.
5. Create `SBI`, `ICICI`, and `PNB` classes that extend `RBI`.
6. Override the `rateOfInterest()` method in each bank class with its respective interest rate.
7. Create the `BankProgram` class with the `main()` method.
8. Create a `Scanner` object to accept input.
9. Read the customer name and customer ID.
10. Create a `Customer` object using the entered details.

11. Read the account number and balance.
12. Create an **Account** object using the entered details.
13. Read the bank name.
14. Create an **RBI** reference variable.
15. Assign the appropriate bank object (**RBI**, **SBI**, **ICICI**, or **PNB**) to the **RBI** reference based on the bank name.
16. Call **rateOfInterest()** using the **RBI** reference.
17. Display the customer details, account details, and applicable interest rate.
18. Close the Scanner.
19. End the program.

Algorithm

1. **Start**
2. Create a **Customer** class with **name** and **customerId**.
3. Create an **Account** class with **accountNumber** and **balance**.
4. Create an **RBI** class with **rateOfInterest()** returning **4%**.
5. Derive **SBI**, **ICICI**, and **PNB** from **RBI**.
6. Override **rateOfInterest()** in each derived class.
7. Read customer name and customer ID.

8. Create a **Customer** object.
9. Read account number and balance.
10. Create an **Account** object.
11. Read the bank name.
12. Create an **RBI** reference.
13. If the bank name is **RBI**, assign an **RBI** object.
14. If the bank name is **SBI**, assign an **SBI** object.
15. If the bank name is **ICICI**, assign an **ICICI** object.
16. If the bank name is **PNB**, assign a **PNB** object.
17. If the bank name is invalid, display **"Invalid bank name"** and terminate.
18. Call **rateOfInterest()** through the **RBI** reference.
19. Display the applicable interest rate.
20. **Stop.**



```
File Edit Selection View Go Run Terminal Help week7
EXPLORER
WEEK7
BankProgram.java
BankProgram.java
60 public class BankProgram {
83
84     System.out.print(s: "Enter the Bank name to find
85     String bankName = sc.nextLine();
86
87     RBI bank;
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\venik\Documents\Records\week7> ^C
PS C:\Users\venik\Documents\Records\week7>
PS C:\Users\venik\Documents\Records\week7> c.; cd 'c:\Users\venik\Docu
a.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\venik\
0bab591dfa\redhat.java\jdt_ws\week7_e0fb52b2\bin' 'BankProgram'
Enter Customer Name: Venika
Enter Customer ID: 10012
Enter Account Number: 25256854
Enter Balance: 250000
Enter the Bank name to find the rate of Interest: SBI
```